

Section Id :	640653106142
Section Number :	6
Section type :	Online
Mandatory or Optional :	Mandatory
Number of Questions :	6
Number of Questions to be attempted :	6
Section Marks :	25
Display Number Panel :	Yes
Section Negative Marks :	0
Group All Questions :	No
Enable Mark as Answered Mark for Review and Clear Response :	No
Section Maximum Duration :	0
Section Minimum Duration :	0
Section Time In :	Minutes
Maximum Instruction Time :	0
Sub-Section Number :	1
Sub-Section Id :	640653245475
Question Shuffling Allowed :	No

Question Number : 92 Question Id : 6406531529758 Question Type : MCQ
Correct Marks : 0

Question Label : Multiple Choice Question

THIS IS QUESTION PAPER FOR THE SUBJECT "DEGREE LEVEL : GAME THEORY AND STRATEGY (COMPUTER BASED EXAM)"

ARE YOU SURE YOU HAVE TO WRITE EXAM FOR THIS SUBJECT?
CROSS CHECK YOUR HALL TICKET TO CONFIRM THE SUBJECTS TO BE WRITTEN.

(IF IT IS NOT THE CORRECT SUBJECT, PLS CHECK THE SECTION AT THE TOP FOR THE SUBJECTS REGISTERED BY YOU)

Options :

6406535121085. ✓ YES

6406535121086. ✗ NO

Sub-Section Number :	2
Sub-Section Id :	640653245476
Question Shuffling Allowed :	No

Question Id : 6406531529759 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix
Question Numbers : (93 to 94)
Question Label : Comprehension

Consider a duopoly game in which two firms simultaneously and independently select prices that are greater than or equal to zero.

Denote firm 1's price as P_1 and firm 2's price as P_2 .

After the prices are set, consumers demand $10 - P_1 + P_2$ units of the good that firm 1 produces, and they demand $10 - P_2 + P_1$ units of the good that firm 2 produces. Assume that each firm produces at zero cost, so firm i 's payoff (profit) is $(10 - P_i + P_j)P_i$, where P_i is firm i 's price and P_j is the other firm's price. Answer the given sub questions based on this information:

Sub questions

Question Number : 93 Question Id : 6406531529760 Question Type : MCQ

Correct Marks : 2

Question Label : Multiple Choice Question

$BR_i(P_j)$ will be

Options :

6406535121087. ✖ $5 + \frac{P_j}{3}$

6406535121088. ✖ $5 + \frac{2P_j}{3}$

6406535121089. ✔ $5 + \frac{P_j}{2}$

6406535121090. ✖ $5 + P_j$

Question Number : 94 Question Id : 6406531529761 Question Type : MCQ

Correct Marks : 2

Question Label : Multiple Choice Question

Choose the incorrect statement

Options :

6406535121091. ✖ This game is a symmetric game

6406535121092. ✖ This game has a symmetric Nash equilibrium

6406535121093. ✔ Under the Nash equilibrium each player chooses $P=15$

6406535121094. ✖ This game has unique Nash equilibrium

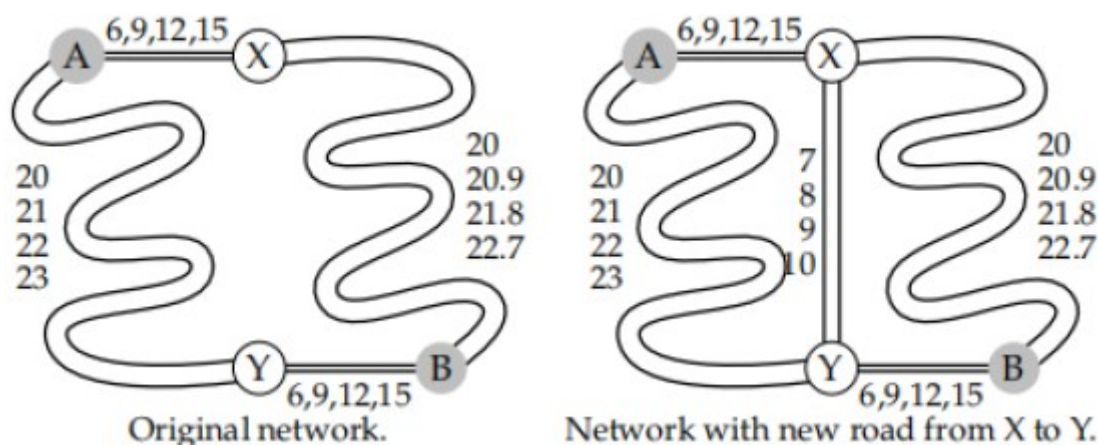
Question Id : 6406531529762 Question Type : COMPREHENSION Sub Question Shuffling

Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

Question Numbers : (95 to 96)

Question Label : Comprehension

(Choosing a route) Four people must drive from A to B at the same time. Two routes are available, one via X and one via Y (Refer to the left panel of following Figure). The roads from A to X, and from Y to B are both short and narrow; in each case, one car takes 6 minutes, and each additional car increases the travel time per car by 3 minutes. (If two cars drive from A to X, for example, each car takes 9 minutes.) The roads from A to Y, and from X to B are long and wide; on A to Y one car takes 20 minutes, and each additional car increases the travel time per car by 1 minute; on X to B one car takes 20 minutes, and each additional car increases the travel time per car by 0.9 minutes. Formulate this situation as a strategic game and answer the given subquestions.



Sub questions

Question Number : 95 Question Id : 6406531529763 Question Type : MCQ

Correct Marks : 2

Question Label : Multiple Choice Question

Choose the correct alternative

Options :

6406535121095. ✖ In Nash equilibrium two people take each route

6406535121096. ✖ For the NE action profile, each person's travel time is either 29.9 or 30 minutes

6406535121097. ✔ Both In Nash equilibrium two people take each route & For the NE action profile, each person's travel time is either 29.9 or 30 minutes

6406535121098. ✖ None

Question Number : 96 Question Id : 6406531529764 Question Type : MCQ

Correct Marks : 2

Question Label : Multiple Choice Question

Now suppose that a relatively short, wide road is built from X to Y, giving each person four options for travel from A to B: A-X-B, A-Y-B, A-X-Y-B, and A-Y-X-B (As shown in the right panel of the given figure). Assume that a person who takes A-X-Y-B travels the A-X portion at the same time as someone who takes A-X-B, and the Y-B portion at the same time as someone who takes A-Y-B. (Think of there being constant flows of traffic.) On the road between X and Y, one car takes 7 minutes and each additional car increases the travel time per car by 1 minute. Consider the Nash

equilibrium in this new situation.

Choose the correct alternative

Options :

6406535121099. ✖ In any Nash equilibrium, two persons take A-X-B, one person takes A-X-Y-B, and one person takes A-Y-B

6406535121100. ✔ In equilibrium profile, every person's travel time increases compared to the case when new road was not there

6406535121101. ✖ Both In any Nash equilibrium, two persons take A-X-B, one person takes A-X-Y-B, and one person takes A-Y-B & In equilibrium profile, every person's travel time increases compared to the case when new road was not there

6406535121102. ✖ None

Question Id : 6406531529777 Question Type : COMPREHENSION Sub Question Shuffling

Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

Question Numbers : (97 to 98)

Question Label : Comprehension

An item is up for auction. Player 1 values the item at 3 while player 2 values the item at 5. Each player can bid either 0, 1 or 2. If player i bids more than player j then i wins the good and pays his bid, while the loser does not pay. If both players bid the same amount, then a coin is tossed to determine who the winner is, who gets the good and pays his bid while the loser pays nothing. Formulate the game in matrix form and answer the given subquestions.

Sub questions

Question Number : 97 Question Id : 6406531529778 Question Type : MCQ

Correct Marks : 2

Question Label : Multiple Choice Question

Which player does not have strictly dominated strategy?

Options :

6406535121140. ✔ Player 1

6406535121141. ✖ Player 2

6406535121142. ✖ None

Question Number : 98 Question Id : 6406531529779 Question Type : MCQ

Correct Marks : 2

Question Label : Multiple Choice Question

Which strategies survive Iterated Elimination of Strictly Dominated Strategy?

Options :

6406535121143. ✖ (0, 0)

6406535121144. ✖ (1, 1)

6406535121145. ✔ (2, 2)

6406535121146. ✖ None

Question Id : 6406531529765 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Question Numbers : (99 to 100)

Question Label : Comprehension

Consider the following normal form game and answer the given sub questions:

		Player 2	
		Left	Right
Player 1	Top	8,2	1,6
	Bottom	3,5	4,1

Sub questions

Question Number : 99 Question Id : 6406531529766 Question Type : MCQ Correct Marks : 2

Question Label : Multiple Choice Question

This game has

Options :

6406535121103. ✖ A unique Nash equilibrium in pure strategy
6406535121104. ✔ A unique Nash equilibrium in mixed strategy
6406535121105. ✖ Both are true
6406535121106. ✖ None is true

Question Number : 100 Question Id : 6406531529767 Question Type : SA Correct Marks : 3

Question Label : Short Answer Question

If under mixed strategy equilibrium, let p and q denote the probability of player 1's choosing 'Top' and player 2's choosing 'Left', respectively,

value of $\frac{1}{p-q}$ will be _____

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

8

Sub-Section Number : 4

Sub-Section Id : 640653245478

Question Shuffling Allowed : No

Question Id : 6406531529768 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

Question Numbers : (101 to 108)

Question Label : Comprehension

Consider a two-player game in which player 1 can choose A or B. The game ends if he chooses A while it continues to player 2 if he chooses B. Player 2 can either choose C or D, with the game ending after C and continuing again with player 1 after D. Player 1 then can choose E or F, and then the game ends after each of these choices.

Imagine that the payoffs following choice A by player 1 are (2, 0), following C by player 2 are (3, 1), following E by player 1 are (0, 0) and following F by player 1 are (1, 2)

Model this as an extensive form game and answer the given subquestions:

Sub questions

Question Number : 101 Question Id : 6406531529769 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

This is a game of

Options :

6406535121108. ✓ Perfect information

6406535121109. ✗ Imperfect information

6406535121110. ✗ None

6406535121111. ✗ Can't be determined

Question Number : 102 Question Id : 6406531529770 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

How many terminal nodes does this game have?

Options :

6406535121112. ✗ 3

6406535121113. ✗ 2

6406535121114. ✓ 4

6406535121115. ✗ 5

Question Number : 103 Question Id : 6406531529771 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

How many information sets does this game have?

Options :

6406535121116. ✖ 2

6406535121117. ✔ 3

6406535121118. ✖ 4

6406535121119. ✖ 1

Question Number : 104 Question Id : 6406531529772 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

How many pure strategies does player 1 have?

Options :

6406535121120. ✖ 1

6406535121121. ✖ 2

6406535121122. ✖ 3

6406535121123. ✔ 4

Question Number : 105 Question Id : 6406531529773 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

How many pure strategies does player 2 have?

Options :

6406535121124. ✖ 1

6406535121125. ✔ 2

6406535121126. ✖ 3

6406535121127. ✖ 4

Question Number : 106 Question Id : 6406531529774 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

How many pure strategy Nash equilibria does this game have?

Options :

6406535121128. ✖ 1

6406535121129. ✖ 2

6406535121130. ✔ 3

6406535121131. ✖ 4

Question Number : 107 Question Id : 6406531529775 Question Type : MCQ

Correct Marks : 1